

NIVELL 1: Exercici 1

La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company"). Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit". Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.

Creació de la taula:

```
1  /* Sprint 3: consulta nivell 1 - exercici 1 - punt 1*/
2  /* crear una taula anomenada "credit_card" */
3
4  CREATE TABLE IF NOT EXISTS credit_card (
5      id VARCHAR(10) PRIMARY KEY,
6      iban VARCHAR(50),
7      pan VARCHAR(50),
8      pin VARCHAR(10),
9      cvv VARCHAR(10),
10     expiring_date VARCHAR(50)
11 );
```

Estableixo les relacions amb *transactions* i *company*

```
/*establir una relació adequada amb les altres dues taules "transaction" */
ALTER TABLE transaction ADD CONSTRAINT fk_credit_card_id FOREIGN KEY (credit_card_id) REFERENCES credit_card(id);
```

El primer cop que executo el codi anterior em dona un error degut a que la taula **crèdit_card** esta buida i no hi ha integritat de dades que es el que es busca amb les claus foranes.

Per aquest motiu he executat el fitxer **datos_introducir_sprint3_credit.sql** i per inserta registres a la nova taula **crèdit_card** i després he tornat a executar el codi anterior per afegir la clau forana a la taula **transaction**. En aquest cas si que s'ha executat sense cap error i ha creat la relació.

```
1  -- Insertamos datos de credit_card
2  INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2938', 'TR301950312213576817638661', '5424465566813633', '3257', '98
3  INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2945', 'D026854763748537475216568689', '5142423821948828', '9080', '
4  INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2952', 'B645IVQL52710525608255', '4556 453 55 5287', '4598', '438', '
5  INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2959', 'CR7242477244335841535', '372461377349375', '3583', '667', '8
6  INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-2966', 'B672LKTQ15627628377363', '448566 886747 7265', '4900', '138', '
--
```

No he creat cap clau forana per a la taula **company** ja que estan relacionades a través de la taula **transaction**.

NIVELL 1: Exercici 2

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar.

The screenshot shows a SQL IDE with a script to update the 'credit_card' table. The script includes comments in Catalan and SQL commands to update the 'iban' field for the record with 'id' 'CcU-2938' and then select the updated record.

```

1 /* Sprint 3: consulta nivell 1 - exercici 1 - punt 2 */
2 /* modificar el compte associat a la targeta de crèdit amb ID CcU-2938 */
3
4 • UPDATE credit_card SET iban = 'TR323456312213576817699999' WHERE id = 'CcU-2938';
5
6 /* mostro el registre modificat */
7 • SELECT iban FROM credit_card WHERE id = 'CcU-2938';

```

The output log shows the execution of the script, including the update and the subsequent select query, with timestamps and durations.

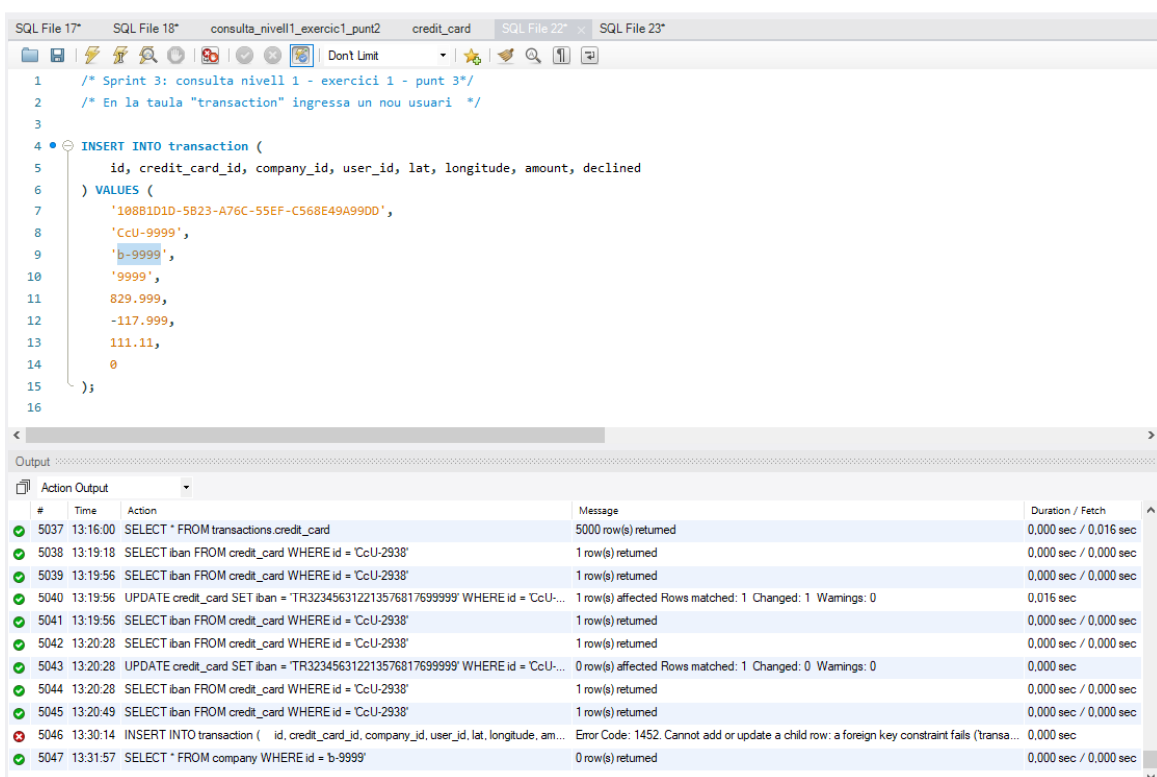
#	Time	Action	Message	Duration / Fetch
5033	12:01:26	SELECT * FROM transactions.credit_card	5000 row(s) returned	0,000 sec / 0,015 sec
5034	12:01:36	ALTER TABLE transaction ADD CONSTRAINT fk_credit_card_id FOREIGN KEY (credit_card_id) REFERENCES credit_card (id)	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	2,656 sec
5035	13:09:08	SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, COLUMN_KEY, COLUMN_COMMENT FROM information_schema.columns WHERE table_name = 'credit_card'	9 row(s) returned	0,016 sec / 0,000 sec
5036	13:09:12	SELECT k.COLUMN_NAME, k.REFERENCED_TABLE_NAME, k.REFERENCED_COLUMN_NAME FROM information_schema.key_column_usage WHERE table_name = 'credit_card'	2 row(s) returned	0,000 sec / 0,000 sec
5037	13:16:00	SELECT * FROM transactions.credit_card	5000 row(s) returned	0,000 sec / 0,016 sec
5038	13:19:18	SELECT iban FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned	0,000 sec / 0,000 sec
5039	13:19:56	SELECT iban FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned	0,000 sec / 0,000 sec
5040	13:19:56	UPDATE credit_card SET iban = 'TR323456312213576817699999' WHERE id = 'CcU-2938'	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0,016 sec
5041	13:19:56	SELECT iban FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned	0,000 sec / 0,000 sec
5042	13:20:28	SELECT iban FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned	0,000 sec / 0,000 sec
5043	13:20:28	UPDATE credit_card SET iban = 'TR323456312213576817699999' WHERE id = 'CcU-2938'	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0,000 sec
5044	13:20:28	SELECT iban FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned	0,000 sec / 0,000 sec

NIVELL 1: Exercici 3

En la taula "transaction" ingressa un nou usuari amb la següent informació:

Id	108B1D1D-5B23-A76C-55EF-C568E49A99DD
credit_card_id	CcU-9999
company_id	b-9999
user_id	9999

lat	829.999
longitude	-117.999
amount	111.11
declined	0



```
1  /* Sprint 3: consulta nivell 1 - exercici 1 - punt 3 */
2  /* En la taula "transaction" ingressa un nou usuari */
3
4  INSERT INTO transaction (
5      id, credit_card_id, company_id, user_id, lat, longitude, amount, declined
6  ) VALUES (
7      '108B1D1D-5B23-A76C-55EF-C568E49A990D',
8      'CcU-99999',
9      'b-9999',
10     '9999',
11     829.999,
12     -117.999,
13     111.11,
14     0
15 );
16
```

Output:

#	Time	Action	Message	Duration / Fetch
5037	13:16:00	SELECT * FROM transactions.credit_card	5000 row(s) returned	0.000 sec / 0.016 sec
5038	13:19:18	SELECT iban FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned	0.000 sec / 0.000 sec
5039	13:19:56	SELECT iban FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned	0.000 sec / 0.000 sec
5040	13:19:56	UPDATE credit_card SET iban = 'TR323456312213576817699999' WHERE id = 'CcU-2938'	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
5041	13:19:56	SELECT iban FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned	0.000 sec / 0.000 sec
5042	13:20:28	SELECT iban FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned	0.000 sec / 0.000 sec
5043	13:20:28	UPDATE credit_card SET iban = 'TR323456312213576817699999' WHERE id = 'CcU-2938'	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec
5044	13:20:28	SELECT iban FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned	0.000 sec / 0.000 sec
5045	13:20:49	SELECT iban FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned	0.000 sec / 0.000 sec
5046	13:30:14	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, am...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (transa...	0.000 sec
5047	13:31:57	SELECT * FROM company WHERE id = 'b-9999'	0 row(s) returned	0.000 sec / 0.000 sec

No es pot insertar aquest registre perquè dona un error de restricció de clau forana sobre el camp **company_id** bàsicament perquè aquesta companyia no existeix a la taula **company**. D'aquesta manera, evitant la inserció d'aquest registre, conservem la integritat de dades.

Una manera de solucionar-lo i poder insertar el registre anterior en la taula **transaction** seria introduir un registre (INSERT) a la taula **company** amb el **company_id** = 'b-999' i més endavant quan tinguéssim la resta de les dades fer un UPDATE a la taula **company** sobre **company_id** = 'b-999'.

NIVELL 1: Exercici 4

Des de recursos humans et solliciten eliminar la columna "pan" de la taula credit_card. Recordar mostrar el canvi realitzat.

```

1  /* Sprint 3: consulta nivell 1 - exercici 1 - punt 4 */
2  /* eliminar la columna "pan" de la taula credit_card */
3
4  • ALTER TABLE credit_card DROP column pan;
5
6  /* mostro els camps de la taula credit_card */
7
8  • SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, COLUMN_KEY, COLUMN_DEFAULT
9  FROM INFORMATION_SCHEMA.COLUMNS
10 WHERE TABLE_SCHEMA = 'transactions'
11 AND TABLE_NAME = 'credit_card';

```

COLUMN_NAME	DATA_TYPE	IS_NULLABLE	COLUMN_KEY	COLUMN_DEFAULT
cvv	varchar	YES		NULL
expiring_date	varchar	YES		NULL
iban	varchar	YES		NULL
id	varchar	NO	PRI	NULL
pin	varchar	YES		NULL

#	Time	Action	Message	Duration / Fetch
5045	13:20:49	SELECT iban FROM credit_card WHERE id = 'CcU-2938'	1 row(s) returned	0.000 sec / 0.000 sec
5046	13:30:14	INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, am...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (transa...	0.000 sec
5047	13:31:57	SELECT * FROM company WHERE id = b-9999'	0 row(s) returned	0.000 sec / 0.000 sec
5048	13:41:06	SELECT * FROM INFORMATION_SCHEMA.COLUMNS WHERE TABLE_SCHEMA = t...	6 row(s) returned	0.000 sec / 0.000 sec
5049	13:41:39	SELECT COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE TAB...	6 row(s) returned	0.000 sec / 0.000 sec
5050	13:42:18	SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, COLUMN_KEY, COLUMN_...	6 row(s) returned	0.000 sec / 0.000 sec
5051	13:43:19	ALTER TABLE credit_card DROP column pan	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
5052	13:43:19	SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, COLUMN_KEY, COLUMN_...	5 row(s) returned	0.000 sec / 0.000 sec

També d'una forma més senzilla podríem verificar que no existeix el camp amb el següent codi:

```

1 • select * from credit_card limit 1

```

id	iban	pin	cvv	expiring_date	fecha_actual
CcS-4857	XX4857591835292505850771	1819	467	09/27/25	2025-07-01

#	Time	Action	Message	Duration / Fetch
1	14:00:14	select * from credit_card limit 1	1 row(s) returned	0.000 sec / 0.000 sec

NIVELL 2: Exercici 1

Elimina de la taula **transaction** el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.

The screenshot shows a SQL IDE with a script editor and a results pane. The script editor contains the following SQL code:

```

1 /* Sprint 3: consulta nivell 2 - exercici 1 */
2 /* Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades */
3
4 DELETE FROM transaction WHERE id='000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';
5
6 /* mostro el registre pre comprobar que ha estat borrat */
7
8 SELECT * FROM transaction WHERE id='000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';

```

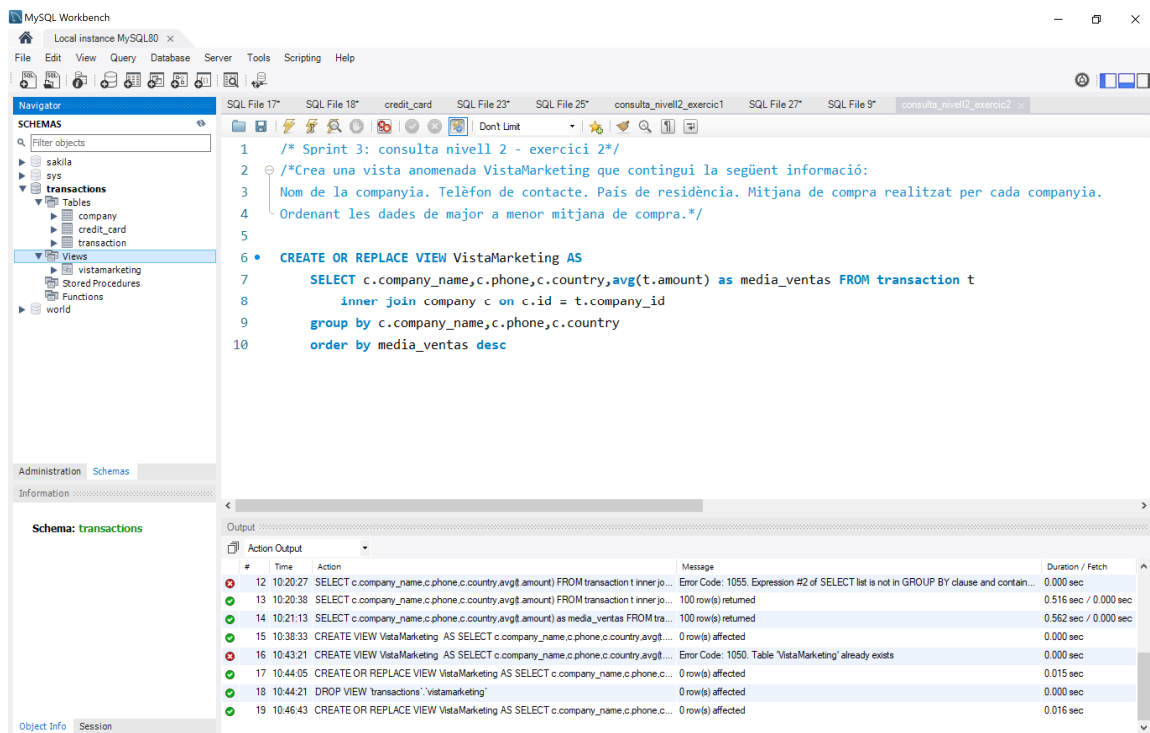
The results pane shows the execution output of the SQL script. The output is as follows:

#	Time	Action	Message	Duration / Fetch
5048	13:41:06	SELECT * FROM INFORMATION_SCHEMA.COLUMNS WHERE TABLE_SCHEMA = 't...	6 row(s) returned	0.000 sec / 0.000 sec
5049	13:41:39	SELECT COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE TAB...	6 row(s) returned	0.000 sec / 0.000 sec
5050	13:42:18	SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, COLUMN_KEY, COLUMN...	6 row(s) returned	0.000 sec / 0.000 sec
5051	13:43:19	ALTER TABLE credit_card DROP column pan	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
5052	13:43:19	SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, COLUMN_KEY, COLUMN...	5 row(s) returned	0.000 sec / 0.000 sec
5053	13:51:50	SELECT * FROM transaction WHERE id='000447FE-B650-4DCF-85DE-C7ED0EE1CAAD'	1 row(s) returned	0.016 sec / 0.000 sec
5054	13:52:10	DELETE FROM transaction WHERE id='000447FE-B650-4DCF-85DE-C7ED0EE1CAAD'	1 row(s) affected	0.016 sec
5055	13:52:10	SELECT * FROM transaction WHERE id='000447FE-B650-4DCF-85DE-C7ED0EE1CAAD'	0 row(s) returned	0.000 sec / 0.000 sec

The file path at the bottom is: J:\EMMY\Sprint 3\Nivell 2\consulta_nivell2_exercici1.sql

NIVELL 2: Exercici 2

La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions. Serà necessària que creïs una vista anomenada *VistaMarketing* que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the 'sakila' database selected. The main editor window shows a SQL script for creating a view. The script includes comments in Catalan and a SQL statement that creates or replaces a view named 'VistaMarketing'. The view selects company details and the average transaction amount, ordered by the average amount in descending order. The bottom panel shows the 'Output' tab with a log of SQL actions and their results, including some error messages for already existing tables and views.

```
1  /* Sprint 3: consulta nivell 2 - exercici 2*/
2  /*Crea una vista anomenada VistaMarketing que contingui la següent informació:
3  Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia.
4  Ordenant les dades de major a menor mitjana de compra.*/
5
6  CREATE OR REPLACE VIEW VistaMarketing AS
7      SELECT c.company_name,c.phone,c.country,avg(t.amount) as media_ventas FROM transaction t
8          inner join company c on c.id = t.company_id
9      group by c.company_name,c.phone,c.country
10     order by media_ventas desc
```

#	Time	Action	Message	Duration / Fetch
12	10:20:27	SELECT c.company_name,c.phone,c.country,avg(t.amount) FROM transaction t inner jo...	Error Code: 1055. Expression #2 of SELECT list is not in GROUP BY clause and contain...	0.000 sec
13	10:20:38	SELECT c.company_name,c.phone,c.country,avg(t.amount) FROM transaction t inner jo...	100 row(s) returned	0.516 sec / 0.000 sec
14	10:21:13	SELECT c.company_name,c.phone,c.country,avg(t.amount) as media_ventas FROM tra...	100 row(s) returned	0.562 sec / 0.000 sec
15	10:38:33	CREATE VIEW VistaMarketing AS SELECT c.company_name,c.phone,c.country,avg(t...	0 row(s) affected	0.000 sec
16	10:43:21	CREATE VIEW VistaMarketing AS SELECT c.company_name,c.phone,c.country,avg(t...	Error Code: 1050. Table 'VistaMarketing' already exists	0.000 sec
17	10:44:05	CREATE OR REPLACE VIEW VistaMarketing AS SELECT c.company_name,c.phone,c...	0 row(s) affected	0.015 sec
18	10:44:21	DROP VIEW 'sakila`.`vistaMarketing'	0 row(s) affected	0.000 sec
19	10:46:43	CREATE OR REPLACE VIEW VistaMarketing AS SELECT c.company_name,c.phone,c...	0 row(s) affected	0.016 sec

He posat la clàusula *REPLACE* perquè no doni error en el cas de que ja existeixi la vista.

NIVELL 2: Exercici 3

Filtra la vista *VistaMarketing* per a mostrar només les companyies que tenen el seu país de residència en "Germany"

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
1 /* Sprint 3: consulta nivell 2 - exercici 3*/
2 /*Filtra la vista VistaMarketing per a mostrar només les companyies que tenen el seu país de residència en "Germany"*/
3
4 select * from VistaMarketing where country = 'Germany'
```

The Result Grid displays the following data:

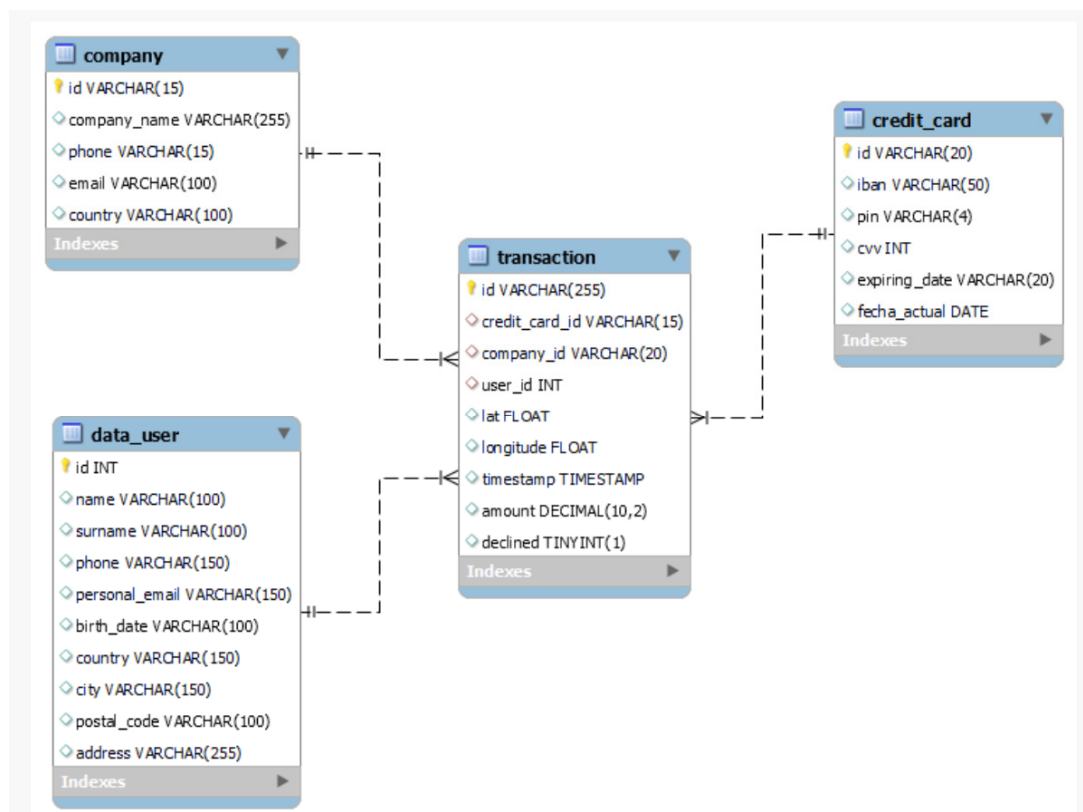
company_name	phone	country	media_ventas
Ac Fermentum Incorporated	06 85 56 52 33	Germany	284.867160
Nunc Interdum Incorporated	05 18 15 48 13	Germany	259.319156
Convalis In Incorporated	06 66 57 29 50	Germany	257.745376
Ac Industries	09 34 65 40 60	Germany	255.147288
Rutrum Non Inc.	02 66 31 61 09	Germany	255.136927
Auctor Mauris Corp.	05 62 87 14 41	Germany	254.765518
Augue Foundation	06 88 43 15 63	Germany	253.505000
Aliquam PC	01 45 73 52 16	Germany	253.136923

The Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
14	10:21:13	SELECT c.company_name,c.phone,c.country,avg(t.amount) as media_ventas FROM tra...	100 row(s) returned	0.562 sec / 0.000 sec
15	10:38:33	CREATE VIEW VistaMarketing AS SELECT c.company_name,c.phone,c.country,avg(t...	0 row(s) affected	0.000 sec
16	10:43:21	CREATE VIEW VistaMarketing AS SELECT c.company_name,c.phone,c.country,avg(t...	Error Code: 1050. Table 'VistaMarketing' already exists	0.000 sec
17	10:44:05	CREATE OR REPLACE VIEW VistaMarketing AS SELECT c.company_name,c.phone,c...	0 row(s) affected	0.015 sec
18	10:44:05	DROP VIEW `transactions`.`vistamarketing`	0 row(s) affected	0.000 sec
19	10:46:43	CREATE OR REPLACE VIEW VistaMarketing AS SELECT c.company_name,c.phone,c...	0 row(s) affected	0.016 sec
20	10:48:19	select * from VistaMarketing	100 row(s) returned	0.421 sec / 0.000 sec
21	10:48:52	select * from VistaMarketing where country = 'Germany'	8 row(s) returned	0.110 sec / 0.000 sec

NIVELL 3: Exercici 1

La setmana vinent tindràs una nova reunió amb els gerents de màrqueting. Un company del teu equip va realitzar modificacions en la base de dades, però no recorda com les va realitzar. Et demana que l'ajudis a deixar els comandos executats per a obtenir el següent diagrama:



```

1  -- Creamos la tabla user
2
3  • CREATE INDEX idx_user_id ON transaction(user_id);
4
5  • CREATE TABLE IF NOT EXISTS user (
6      id INT PRIMARY KEY,
7      name VARCHAR(100),
8      surname VARCHAR(100),
9      phone VARCHAR(150),
10     email VARCHAR(150),
11     birth_date VARCHAR(100),
12     country VARCHAR(150),
13     city VARCHAR(150),
14     postal_code VARCHAR(100),
15     address VARCHAR(255),
16     FOREIGN KEY(id) REFERENCES transaction(user_id)
17 );
18

```

Output

#	Time	Action	Message	Duration / Fetch
16	10:43:21	CREATE VIEW VistaMarketing AS SELECT c.company_name,c.phone,c.country,avg(t...	Error Code: 1050. Table 'VistaMarketing' already exists	0.000 sec
17	10:44:05	CREATE OR REPLACE VIEW VistaMarketing AS SELECT c.company_name,c.phone,c...	0 row(s) affected	0.015 sec
18	10:44:21	DROP VIEW `transactions`.`vistamarketing`	0 row(s) affected	0.000 sec
19	10:46:43	CREATE OR REPLACE VIEW VistaMarketing AS SELECT c.company_name,c.phone,c...	0 row(s) affected	0.016 sec
20	10:48:19	select * from VistaMarketing	100 row(s) returned	0.421 sec / 0.000 sec
21	10:48:52	select * from VistaMarketing where country = 'Germany'	8 row(s) returned	0.110 sec / 0.000 sec
22	11:27:20	CREATE INDEX idx_user_id ON transaction(user_id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.781 sec
23	11:27:20	CREATE TABLE IF NOT EXISTS user (id INT PRIMARY KEY, name VARCHAR...	0 row(s) affected	0.047 sec

The screenshot shows a SQL IDE with a script containing 498 lines of SQL code. Each line is an `INSERT INTO user` statement with columns `(id, name, surname, phone, email, birth_date, country, city, postal_code, address)` and a corresponding `VALUES` clause. The values are a mix of random strings and some recognizable words. The IDE interface includes a toolbar at the top with icons for file operations, a query editor with a 'Don't Limit' dropdown, and an 'Output' window at the bottom. The 'Output' window is set to 'Action Output' and displays the execution results of the script, showing that all 498 insertions were successful, each affecting 1 row.

```

477 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('750', "Psdawn", "Bdpxmlra", "
478 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('751', "Kzalnk", "Xgptzboy", "
479 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('752', "Sdfnju", "Jjxmsgir", "
480 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('755', "Knplaq", "Fjedixlt", "
481 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('756', "Toogtv", "Euspyziq", "
482 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('757', "Qfvræ", "Zazmrvt", "
483 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('758', "Hrgwyd", "Dtvtnbgi", "
484 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('759', "Qqgahw", "Dwlebbdb", "
485 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('760', "Vegtl", "Dfdexgrq", "
486 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('761', "Bmkqxf", "Xlalusq", "
487 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('762', "Clofnx", "Kmvkvyk", "
488 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('763', "Eaagdz", "Vkhuruplq", "
489 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('764', "Gzbqib", "Dvuhsosg", "
490 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ('765', "Zuctzi", "Sotbkzki", "

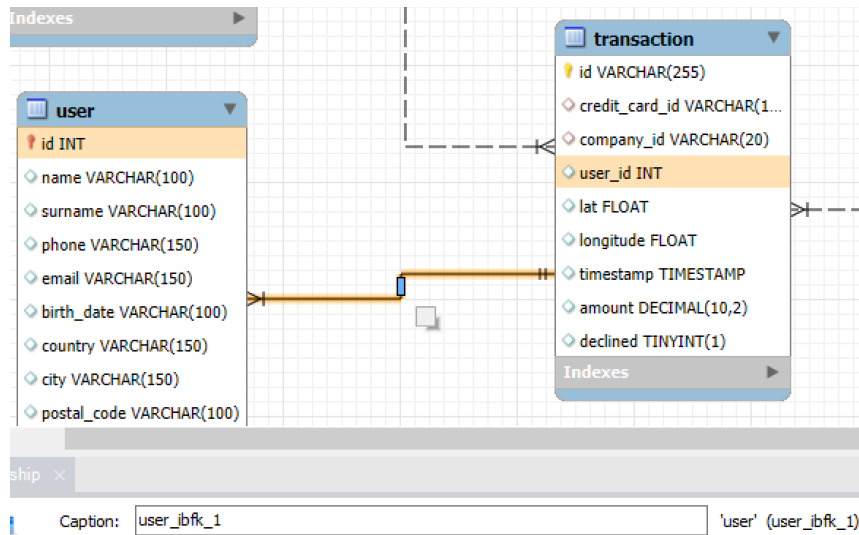
```

Output

#	Time	Action	Message	Duration / Fetch
✓ 5327	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.000 sec
✓ 5328	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.000 sec
✓ 5329	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.015 sec
✓ 5330	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.000 sec
✓ 5331	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.000 sec
✓ 5332	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.016 sec
✓ 5333	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.000 sec

Per rectificar-ho primer eliminarem la restricció i després la crearem de nou.

Per trobar el nom (**user_ibfk_1**) de la restricció ho puc fer visualitzant el esquema com mostra la següent imatge en el *caption*:



3) Un cop trobat el nom/etiqueta de la restricció, la eliminem la i la creem de nou.

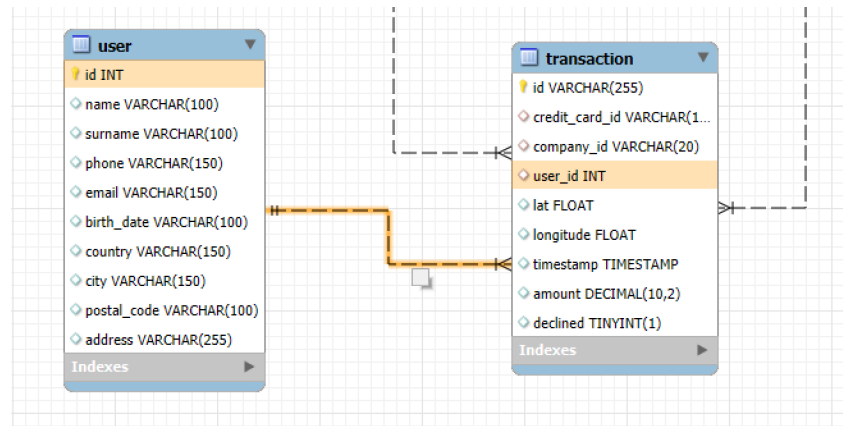
```

1  /* Sprint 3: consulta nivell 3 - exercici 1 */
2
3  /* Eliminem la relació */
4  ALTER TABLE `user`
5  DROP FOREIGN KEY `user_ibfk_1`;
6
7  /* tornem a crear la relació */
8  ALTER TABLE transaction
9  ADD CONSTRAINT fk_transaction_user
10 FOREIGN KEY (user_id)
11 REFERENCES user(id)
12 ON UPDATE CASCADE;
13

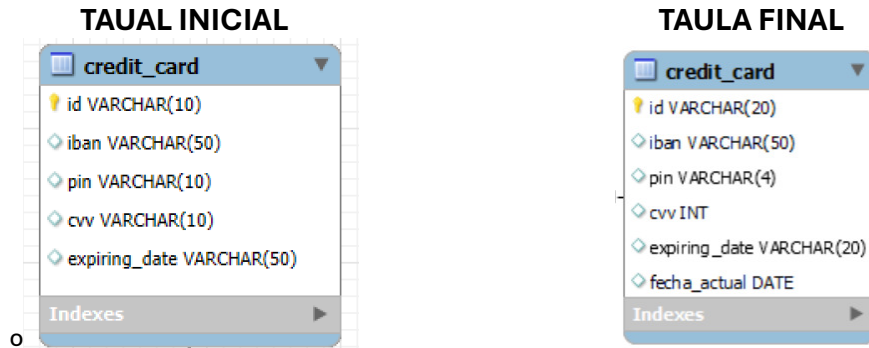
```

#	Time	Action	Message	Duration / Fetch
5325	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.000 sec
5326	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.000 sec
5327	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.000 sec
5328	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.000 sec
5329	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.015 sec
5330	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.000 sec
5331	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.000 sec
5332	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.016 sec
5333	12:48:42	INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_c...	1 row(s) affected	0.000 sec
5334	12:49:21	select count(*) from user	1 row(s) returned	0.016 sec / 0.000 sec
5335	12:50:25	SELECT DISTINCT user_id FROM transaction WHERE user_id NOT IN (SELECT id FR...	0 row(s) returned	0.047 sec / 0.000 sec
5336	12:50:35	SELECT DISTINCT user_id FROM transaction WHERE user_id NOT IN (SELECT id FR...	0 row(s) returned	0.031 sec / 0.000 sec
5337	12:50:36	SELECT DISTINCT user_id FROM transaction WHERE user_id NOT IN (SELECT id FR...	0 row(s) returned	0.032 sec / 0.000 sec
5338	12:50:52	ALTER TABLE transaction ADD CONSTRAINT fk_transaction_user FOREIGN KEY (us...	99999 row(s) affected Records: 99999 Duplicates: 0 Warnings: 0	3.110 sec

Comprovem ara l'esquema:



Hem de fer varies modificacions sobre els camps de la taula *crèdit_card* perquè coincideixin amb els del esquema.



Modificar tamanay id de VARCHAR(10) a VARCHAR(20). En aquest cas el camp *id* de la taula *credit_card* és clau foranea de la taula *transaction*, per aquest motiu el canvi de tamany del camp s'ha d'aplicar a les dues taules perquè han de ser del mateix. Per fer-ho primer haurem d'eliminar la restricció de clau foranea, modificar els camps a les dues taules i tornar a crear la restricció de clau foranea.

Hom farem amb el següent codi:

```

1  /* Sprint 3: consulta nivell 3 - exercici 1 - part 02*/
2  /* canvis a la taula user camp id */
3  /* 1) Eliminar la restricció de la clau forànea temporalment*/
4  • ALTER TABLE transaction
5    DROP FOREIGN KEY fk_credit_card_id;
6
7  /* 2) Canviar la mida de les dues columnes de les dues taules perquè coincideixin*/
8  • ALTER TABLE credit_card
9    MODIFY COLUMN id VARCHAR(20) NOT NULL;
10
11 • ALTER TABLE transaction
12   MODIFY COLUMN credit_card_id VARCHAR(20) NOT NULL;
13
14 /* 3) tornar a crear la clau forànea */
15 • ALTER TABLE transaction
16   ADD CONSTRAINT fk_credit_card_id
17   FOREIGN KEY (credit_card_id)
18   REFERENCES credit_card(id)
19   ON UPDATE CASCADE;

```

#	Time	Action	Message	Duration / Fetch
5353	13:36:53	ALTER TABLE transaction ADD CONSTRAINT fk_transaction_user FOREIGN KEY (us...	99999 row(s) affected Records: 99999 Duplicates: 0 Warnings: 0	3.047 sec
5354	13:39:22	ALTER TABLE credit_card MODIFY COLUMN id VARCHAR(20) NOT NULL	Error Code: 1146. Table 'transactions.credit_card' doesn't exist	0.000 sec
5355	13:39:30	ALTER TABLE credit_card MODIFY COLUMN id VARCHAR(20) NOT NULL	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
5356	13:41:48	ALTER TABLE credit_card DROP FOREIGN KEY fk_credit_card_id	Error Code: 1091. Can't DROP 'fk_credit_card_id': check that column/key exists	0.000 sec
5357	13:42:17	ALTER TABLE transaction DROP FOREIGN KEY fk_credit_card_id	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
5358	13:42:17	ALTER TABLE credit_card MODIFY COLUMN id VARCHAR(20) NOT NULL	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
5359	13:42:17	ALTER TABLE transaction MODIFY COLUMN credit_card_id VARCHAR(20) NOT NULL	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	3.141 sec
5360	13:42:20	ALTER TABLE transaction ADD CONSTRAINT fk_credit_card_id FOREIGN KEY (credi...	99999 row(s) affected Records: 99999 Duplicates: 0 Warnings: 0	3.594 sec

Altres camps a modificar de la taula credit_card:

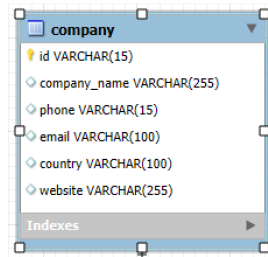
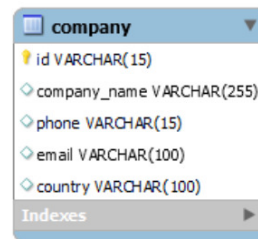
- Modificació pin VARCHAR(10) a pin VARCHAR(4)
- Modificació cvv VARCHAR(10) a cvv INT
- Modificació expiring_date VARCHAR(50) a expiring_date VARCHAR(20)
- Afegir el camp fecha_actual de tipus DATE amb valor per defecte de la data actual.

```

1  /* Sprint 3: consulta nivell 3 - exercici 1 - part 3*/
2  /* canvis a la taula credit_card a diferents camps */
3
4  • ALTER TABLE credit_card
5    MODIFY COLUMN pin VARCHAR(4) NOT NULL;
6
7  • ALTER TABLE credit_card
8    MODIFY COLUMN cvv INT NOT NULL;
9
10 • ALTER TABLE credit_card
11   MODIFY COLUMN expiring_date VARCHAR(20) NOT NULL;
12
13 • ALTER TABLE credit_card
14   ADD COLUMN fecha_actual DATE DEFAULT (CURRENT_DATE);
15
16 /* CURRENT_DATE és estàndard ANSI SQL i funciona a MySQL per assignar la data actual del sistema. */

```

Ara hem de fer una modificació sobre la taula company perquè coincideixin amb els del esquema.

TAUAL INICIAL**TAUAL FINAL**

Eliminar el camp *website* VARCHAR(255)

```
1  /* Sprint 3: consulta nivell 3 - exercici 1 - part 4*/  
2  /* Eliminar el camp website de la taula company */  
3  • ALTER TABLE company  
4    DROP COLUMN website;  
5
```

Output**Action Output**

#	Time	Action
1	14:10:32	ALTER TABLE company DROP COLUMN website

Message

0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

NIVELL 3: Exercici 2

L'empresa també us demana crear una vista anomenada "InformeTecnico" que contingui la següent informació:

- ID de la transacció
- Nom de l'usuari/ària
- Cognom de l'usuari/ària
- IBAN de la targeta de crèdit usada.
- Nom de la companyia de la transacció realitzada.
- Assegureu-vos d'incloure informació rellevant de les taules que coneixereu i utilitzeu àlies per canviar de nom columnes segons calgui.

Mostra els resultats de la vista, ordena els resultats de forma descendent en funció de la variable ID de transacció.

The screenshot shows a SQL IDE interface with a Navigator on the left, a SQL editor in the center, and an Output window at the bottom.

Navigator: Shows a tree structure with 'sakila' and 'sys' schemas. Under 'sys', there are 'tables' and 'views'. The 'views' folder is expanded, showing 'informetecnico', 'vistamarketing', 'Stored Procedures', and 'Functions'. The 'Schema: transactions' is selected.

SQL Editor: Contains the following SQL code:

```

1  /* Sprint 3: consulta nivell 3 - exercici 2*/
2  /* crear una vista anomenada "InformeTecnico" */
3
4  CREATE OR REPLACE VIEW InformeTecnico AS
5      select t.id as Transaccio, u.name, u.surname, cc.iban, c.company_name as Companyia from transaction t
6          inner join user u on u.id = t.user_id
7          inner join credit_card cc on cc.id = t.credit_card_id
8          inner join company c on c.id = t.company_id
9      order by t.id;

```

Output: The Output window shows the execution results of the SQL script. It includes a table with columns: #, Time, Action, Message, and Duration / Fetch.

#	Time	Action	Message	Duration / Fetch
1	10:36:43	SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, COLUMN_KEY, COLUMN_...	9 row(s) returned	0.000 sec / 0.000 sec
2	10:41:37	SELECT k.COLUMN_NAME, k.REFERENCED_TABLE_NAME, k.REFERENCED_C...	3 row(s) returned	0.000 sec / 0.000 sec
3	10:42:37	SHOW COLUMNS FROM user LIKE 'id'	1 row(s) returned	0.000 sec / 0.000 sec
4	10:42:37	SHOW INDEX FROM user WHERE Key_name = 'PRIMARY'	1 row(s) returned	0.000 sec / 0.000 sec
5	10:42:59	SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, COLUMN_KEY, COLUMN_DE...	6 row(s) returned	0.000 sec / 0.000 sec
6	10:46:21	select * from transaction t inner join user u on u.id = t.user_id inner join credit_card cc on c...	99999 row(s) returned	0.031 sec / 2.125 sec
7	10:50:46	select t.id, u.name, u.surname, cc.iban, c.company_name from transaction t inner join use...	99999 row(s) returned	0.781 sec / 0.062 sec
8	10:51:53	select t.id, u.name, u.surname, cc.iban, c.company_name from transaction t inner join use...	99999 row(s) returned	0.735 sec / 0.062 sec
9	10:52:26	select t.id as Transaccio, u.name, u.surname, cc.iban, c.company_name from transaction ...	99999 row(s) returned	0.719 sec / 0.047 sec
10	10:52:50	select t.id as Transaccio, u.name, u.surname, cc.iban, c.company_name as Companyia fr...	99999 row(s) returned	1.032 sec / 0.156 sec
11	10:55:00	CREATE OR REPLACE VIEW InformeTecnico AS select t.id as Transaccio, u.name, u.su...	0 row(s) affected	0.016 sec